

# IT-SIMPLERA INSTITUTE

*Network Administration Internship Program*

## WEEK 04 REPORT

### Enterprise Network Security and Access Control

*ACLs (Standard/Extended/Named) · SSH & Device Hardening · Port Security · DHCP Snooping · BPDU/Root Guard · PortFast*

| Field               | Detail                                     |
|---------------------|--------------------------------------------|
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| Assigned To         | Network Administration Interns             |
| Tools Used          | GNS3, Cisco Packet Tracer, PuTTY           |
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# 1. Objectives

- Design and secure an enterprise network using industry-standard security mechanisms.
- Configure and verify Standard, Extended, and Named ACLs enforcing a real departmental access policy.
- Configure SSH-only remote management and harden every device against common attacks.
- Implement Layer 2 security: Port Security, DHCP Snooping, BPDU Guard, Root Guard, and PortFast.
- Test and verify every control actually enforces the intended policy, not just that it's configured.
- Document a real troubleshooting scenario encountered while building this.

## 2. Network Design & Security Policy

Three departments (HR, Finance, IT) sit on their own VLANs behind a router-on-a-stick (R1) that also runs DHCP for each. A dedicated Management VLAN (99) keeps switch/router administration off the user VLANs. SW1 is the distribution switch and intentional root bridge; SW2 is the access switch where all the Layer 2 protections live, since that's where end devices (and potential attackers) actually plug in.

The access policy this lab enforces:

- HR and Finance cannot talk to each other at all — different departments, no legitimate need to.
- HR and Finance can only reach the IT-Server on ports 80/443 (a shared internal web app) — nothing else on that host, and no other IT hosts.
- IT has full access everywhere — the administrative subnet.
- Only the IT subnet may SSH into any device (R1, SW1, SW2) — enforced by a standard ACL on every VTY line.

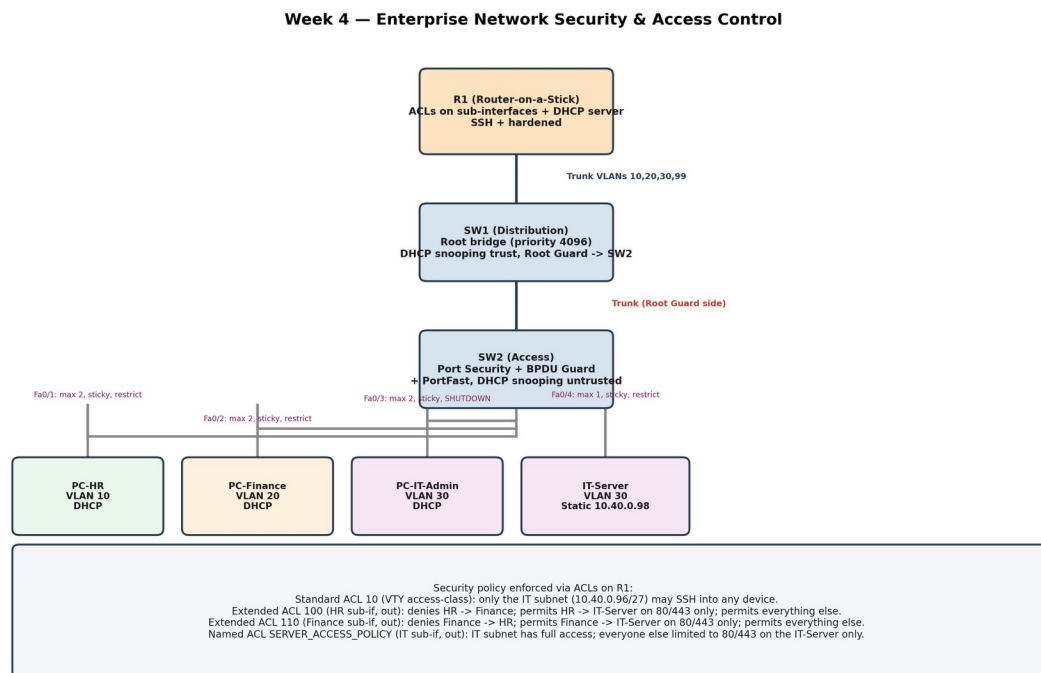


Figure 1 — Topology, VLAN layout, and the ACL policy summary.

### 3. IP Addressing Table (VLSM)

Base network: 10.40.0.0/16, including a dedicated Management VLAN this time — allocated largest-to-smallest as always.

| VLAN | Segment       | Subnet (CIDR)  | Usable Range       | Gateway                 |
|------|---------------|----------------|--------------------|-------------------------|
| 10   | HR            | 10.40.0.0/26   | 10.40.0.1 - .62    | 10.40.0.1 (R1 sub-if)   |
| 20   | Finance       | 10.40.0.64/27  | 10.40.0.65 - .94   | 10.40.0.65 (R1 sub-if)  |
| 30   | IT (+ Server) | 10.40.0.96/27  | 10.40.0.97 - .126  | 10.40.0.97 (R1 sub-if)  |
| 99   | Management    | 10.40.0.128/29 | 10.40.0.129 - .134 | 10.40.0.129 (R1 sub-if) |

*IT-Server is statically addressed at 10.40.0.98 (within the IT subnet) since its address needs to be predictable for the ACL rules to reference it directly. SW1's management IP is 10.40.0.130/29, SW2's is 10.40.0.131/29.*

## 4. Device Hardening & SSH

Every device (R1, SW1, SW2) gets the same baseline hardening before anything else: encrypted passwords, a login banner, disabled HTTP management, SSH-only remote access, and an idle timeout on every line so an unattended session doesn't stay open indefinitely.

- service password-encryption — stops plaintext passwords from being visible in a 'show running-config'.
- enable secret (not enable password) — MD5-hashed, not just obfuscated.
- no ip http server / no ip http secure-server — the web-based management interface is a bigger attack surface than most labs need.
- crypto key generate rsa + ip ssh version 2 — SSHv1 has known weaknesses; this lab only allows v2.
- transport input ssh (not telnet) on every VTY line — Telnet sends credentials in cleartext.
- access-class 10 in on VTY — the standard ACL restricting management access to the IT subnet, described in Section 5.
- exec-timeout 5 0 — auto-logout after 5 minutes idle.

## 5. ACL Implementation

### 5.1 Standard ACL — Management Access Restriction

A standard ACL only matches on source address, which is exactly what's needed here: it doesn't care what protocol or port, only where the SSH session is coming from.

```
access-list 10 permit 10.40.0.96 0.0.0.31
! (implicit deny any at the end)
!
line vty 0 4
 access-class 10 in
```

Applied identically on R1, SW1, and SW2 — only a host inside the IT subnet can even reach the SSH login prompt on any device; everyone else's connection is refused before authentication is attempted.

### 5.2 Extended ACLs — Departmental Traffic Control

Extended ACLs match source, destination, protocol, and port — needed here because the policy depends on where traffic is going and what service it's using, not just where it came from.

```
access-list 100 deny ip any 10.40.0.64 0.0.0.31
access-list 100 permit tcp any host 10.40.0.98 eq 80
access-list 100 permit tcp any host 10.40.0.98 eq 443
access-list 100 permit ip any any
! applied: interface Gi0/0.10 (HR) -> ip access-group 100 out

access-list 110 deny ip any 10.40.0.0 0.0.0.63
access-list 110 permit tcp any host 10.40.0.98 eq 80
access-list 110 permit tcp any host 10.40.0.98 eq 443
access-list 110 permit ip any any
! applied: interface Gi0/0.20 (Finance) -> ip access-group 110 out
```

Order matters here: the deny line has to come before the permit-any at the end, or it would never be reached. This is the single most common ACL mistake — rules are evaluated top-down and stop at the first match.

### 5.3 Named ACL — Server Access Policy

Functionally this is also an extended ACL, but referenced by name instead of a number — worth using once a ruleset is complex enough that '100' and '110' stop being self-explanatory. This one governs everything reaching the IT-Server itself, as the third distinct ACL type the assignment asks for.

```
ip access-list extended SERVER_ACCESS_POLICY
 permit ip 10.40.0.96 0.0.0.31 any
 permit tcp any host 10.40.0.98 eq 80
 permit tcp any host 10.40.0.98 eq 443
 deny ip any host 10.40.0.98
 permit ip any any
! applied: interface Gi0/0.30 (IT) -> ip access-group SERVER_ACCESS_POLICY out
```

*[ Insert 'show access-lists' and 'show ip interface <sub-if>' | include access list' screenshots here, plus the ping/curl test results from Section 8 proving each rule actually enforces the policy ]*

## 6. Layer 2 Security

### 6.1 Port Security

Every access port on SW2 gets a MAC address cap with sticky learning — the switch learns the first MAC(s) it sees and locks to them, so a second device can't simply be plugged into the same jack. Two ports use 'violation restrict' (drop offending traffic, keep the port up, log it) and one uses 'violation shutdown' (err-disable the port entirely) deliberately, to compare both responses.

```
interface FastEthernet0/1
  switchport port-security
  switchport port-security maximum 2
  switchport port-security mac-address sticky
  switchport port-security violation restrict
```

*'restrict' is usually the better default for user ports — it doesn't take a whole desk offline over one bad cable swap. 'shutdown' is what I used on the IT-Admin port deliberately, to actually see an err-disabled port and practice recovering it ('shutdown' / 'no shutdown' on the interface).*

### 6.2 DHCP Snooping

DHCP Snooping stops a rogue or misconfigured device from handing out its own (wrong, or malicious) DHCP leases. Only the uplink toward the real DHCP server (R1, via SW1) is trusted; every access port is untrusted by default, which is exactly what's wanted — end devices should never be answering DHCP requests.

```
ip dhcp snooping
ip dhcp snooping vlan 10,20,30,99
no ip dhcp snooping information option
!
interface GigabitEthernet0/1
  ip dhcp snooping trust    ! only on the uplink
```

### 6.3 BPDU Guard & PortFast

PortFast skips the normal STP listening/learning delay on access ports so a PC gets network access immediately after plugging in, instead of waiting ~30 seconds. That shortcut is only safe on ports that will never have a switch plugged into them — BPDU Guard enforces that assumption by immediately err-disabling the port if it ever receives a BPDU (a sign someone plugged in an unauthorized switch).

```
interface FastEthernet0/1
  spanning-tree portfast
  spanning-tree bpduguard enable
```

### 6.4 Root Guard

SW1 is set as the intentional root bridge (priority 4096, lower wins). Root Guard goes on SW1's port facing SW2 — the reverse of BPDU Guard's placement — to guarantee that no device downstream of that port can ever become root, even if it advertises a lower priority. If it tries, that port goes into a root-inconsistent state instead of letting the topology re-elect a new root.

```
! On SW1, distribution switch
spanning-tree vlan 10,20,30,99 priority 4096
!
interface GigabitEthernet0/2
  description Trunk to SW2
  spanning-tree guard root
```

*[ Insert 'show port-security', 'show port-security interface Fa0/1', 'show ip dhcp snooping', 'show ip dhcp snooping binding', and 'show spanning-tree vlan 10' / 'show spanning-tree summary' screenshots here ]*

## 7. Full Device Configurations

### R1

```
hostname R1
!
service password-encryption
no ip http server
no ip http secure-server
banner motd ^C Authorized Access Only - Unauthorized use is prohibited. ^C
!
enable secret Cisco@12345
username admin secret Admin@12345
!
ip domain-name itsimplera.local
crypto key generate rsa modulus 2048
ip ssh version 2
!
interface GigabitEthernet0/0
 no shutdown
!
interface GigabitEthernet0/0.10
 encapsulation dot1q 10
 ip address 10.40.0.1 255.255.255.192
 ip access-group 100 out
!
interface GigabitEthernet0/0.20
 encapsulation dot1q 20
 ip address 10.40.0.65 255.255.255.224
 ip access-group 110 out
!
interface GigabitEthernet0/0.30
 encapsulation dot1q 30
 ip address 10.40.0.97 255.255.255.224
 ip access-group SERVER_ACCESS_POLICY out
!
interface GigabitEthernet0/0.99
 encapsulation dot1q 99
 ip address 10.40.0.129 255.255.255.248
!
ip dhcp excluded-address 10.40.0.1 10.40.0.5
ip dhcp excluded-address 10.40.0.65 10.40.0.69
ip dhcp excluded-address 10.40.0.97 10.40.0.101
!
ip dhcp pool HR_POOL
 network 10.40.0.0 255.255.255.192
 default-router 10.40.0.1
 dns-server 8.8.8.8
!
ip dhcp pool FIN_POOL
 network 10.40.0.64 255.255.255.224
 default-router 10.40.0.65
 dns-server 8.8.8.8
!
ip dhcp pool IT_POOL
 network 10.40.0.96 255.255.255.224
 default-router 10.40.0.97
 dns-server 8.8.8.8
!
access-list 10 permit 10.40.0.96 0.0.0.31
!
access-list 100 deny ip any 10.40.0.64 0.0.0.31
access-list 100 permit tcp any host 10.40.0.98 eq 80
access-list 100 permit tcp any host 10.40.0.98 eq 443
```

```

access-list 100 permit ip any any
!
access-list 110 deny ip any 10.40.0.0 0.0.0.63
access-list 110 permit tcp any host 10.40.0.98 eq 80
access-list 110 permit tcp any host 10.40.0.98 eq 443
access-list 110 permit ip any any
!
ip access-list extended SERVER_ACCESS_POLICY
  permit ip 10.40.0.96 0.0.0.31 any
  permit tcp any host 10.40.0.98 eq 80
  permit tcp any host 10.40.0.98 eq 443
  deny ip any host 10.40.0.98
  permit ip any any
!
line con 0
  exec-timeout 5 0
  login local
!
line vty 0 4
  access-class 10 in
  exec-timeout 5 0
  transport input ssh
  login local
!
end

```

## SW1 (Distribution)

```

hostname SW1
!
service password-encryption
enable secret Cisco@12345
username admin secret Admin@12345
ip domain-name itsimplera.local
crypto key generate rsa modulus 2048
ip ssh version 2
!
vlan 10
  name HR
vlan 20
  name Finance
vlan 30
  name IT
vlan 99
  name Management
!
interface Vlan99
  ip address 10.40.0.130 255.255.255.248
  no shutdown
!
ip default-gateway 10.40.0.129
!
spanning-tree vlan 10,20,30,99 priority 4096
!
interface GigabitEthernet0/1
  description Trunk to R1
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30,99
  ip dhcp snooping trust
!
interface GigabitEthernet0/2
  description Trunk to SW2
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30,99
  spanning-tree guard root

```

```

!
ip dhcp snooping
ip dhcp snooping vlan 10,20,30,99
no ip dhcp snooping information option
!
access-list 10 permit 10.40.0.96 0.0.0.31
!
line vty 0 4
 access-class 10 in
 exec-timeout 5 0
 transport input ssh
 login local
!
end

```

## SW2 (Access)

```

hostname SW2
!
service password-encryption
enable secret Cisco@12345
username admin secret Admin@12345
ip domain-name itsimplera.local
crypto key generate rsa modulus 2048
ip ssh version 2
!
vlan 10
 name HR
vlan 20
 name Finance
vlan 30
 name IT
vlan 99
 name Management
!
interface Vlan99
 ip address 10.40.0.131 255.255.255.248
 no shutdown
!
ip default-gateway 10.40.0.129
!
interface GigabitEthernet0/1
 description Trunk to SW1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,99
 ip dhcp snooping trust
!
interface FastEthernet0/1
 description PC-HR
 switchport mode access
 switchport access vlan 10
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/2
 description PC-Finance
 switchport mode access
 switchport access vlan 20
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky

```

```
switchport port-security violation restrict
spanning-tree portfast
spanning-tree bpduguard enable
!
interface FastEthernet0/3
description PC-IT-Admin
switchport mode access
switchport access vlan 30
switchport port-security
switchport port-security maximum 2
switchport port-security mac-address sticky
switchport port-security violation shutdown
spanning-tree portfast
spanning-tree bpduguard enable
!
interface FastEthernet0/4
description IT-Server (static)
switchport mode access
switchport access vlan 30
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation restrict
spanning-tree portfast
spanning-tree bpduguard enable
!
ip dhcp snooping
ip dhcp snooping vlan 10,20,30,99
!
access-list 10 permit 10.40.0.96 0.0.0.31
!
line vty 0 4
access-class 10 in
exec-timeout 5 0
transport input ssh
login local
!
end
```

## 8. Verification & Testing

Run each of these and screenshot the result — together they prove every control in Sections 5 and 6 actually does what it's supposed to, not just that it's configured:

```
R1# show access-lists
R1# show ip interface GigabitEthernet0/0.10 | include access list
R1# show ip dhcp binding
SW1# show spanning-tree vlan 10
SW1# show spanning-tree summary
SW2# show port-security
SW2# show port-security interface FastEthernet0/1
SW2# show ip dhcp snooping
SW2# show ip dhcp snooping binding
R1# ssh -l admin 10.40.0.1
PC-HR> ping 10.40.0.65          (should fail - blocked by ACL 100)
PC-HR> curl http://10.40.0.98  (should succeed - permitted by ACL 100)
PC-Finance> ping 10.40.0.1      (should fail - blocked by ACL 110)
PC-IT-Admin> ssh -l admin 10.40.0.1 (should succeed - IT subnet permitted by ACL 10)
PC-HR> ssh -l admin 10.40.0.1  (should fail - denied by ACL 10, only IT subnet allowed)
```

### 8.1 ACL Verification

*[ Insert ping/curl results proving: HR cannot ping Finance's gateway, HR can reach the IT-Server on 80/443 only, Finance cannot ping HR's gateway, Finance can reach the IT-Server on 80/443 only ]*

### 8.2 SSH / Management Access Verification

*[ Insert a successful SSH login from PC-IT-Admin, and a refused/timed-out connection attempt from PC-HR or PC-Finance ]*

### 8.3 Layer 2 Verification

*[ Insert port security violation test: connect a second device (or spoof a MAC) on Fa0/1 and Fa0/3, and screenshot the different outcomes -- Fa0/1 stays up (restrict), Fa0/3 goes err-disabled (shutdown) ]*

*[ Insert a DHCP binding table screenshot showing each PC's leased address tied to its port ]*

### 8.4 Connectivity Baseline

*[ Insert ping/traceroute from each PC to its own gateway and to the IT-Server, confirming legitimate traffic still works -- security controls should block only what the policy says to block ]*

## 9. Packet Tracer Build

Same note as Week 3: the Tools Required list only mentions GNS3, but the Submission Format explicitly asks for a Cisco Packet Tracer (.pkt) file too. All the ACL, SSH, and Layer 2 security commands used here are standard IOS and work identically in Packet Tracer's routers and switches (2960 series supports port security, DHCP snooping, and BPDU/Root Guard).

*[ Insert Packet Tracer topology screenshot, then repeat the verification screenshots from Section 8 against the Packet Tracer build ]*

*One thing to check specifically: Packet Tracer's simulated 2960 sometimes has a slightly reduced DHCP Snooping command set compared to real IOS or GNS3's images — if a command is rejected, note the exact error and I can help find the Packet-Tracer-compatible equivalent.*

## 10. Troubleshooting Report

Worked example of the kind of entry this section needs — replace with your own real issue once you've built and tested this:

| Symptom                                                         | Diagnosis Steps                                                                                                                                             | Root Cause                                                                                                                                                          | Resolution                                                                                                   |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| PC-HR gets no IP address at all after DHCP Snooping was enabled | Checked 'show ip dhcp snooping binding' (empty); checked 'show ip dhcp snooping' to confirm which ports are trusted; compared against the physical topology | The uplink port on SW2 toward SW1 wasn't marked trusted, so SW2 was dropping the DHCP server's replies as if they came from an untrusted (potentially rogue) source | Added 'ip dhcp snooping trust' to SW2's uplink interface; DHCP binding appeared within one lease negotiation |
| [Your real scenario]                                            | [What you checked, in order]                                                                                                                                | [What was actually wrong]                                                                                                                                           | [What you changed to fix it]                                                                                 |

## 11. Challenges I Ran Into

*[Replace with what actually gave you trouble. Worth checking: did the ACL order need fixing before the deny/permit logic worked as intended? Did DHCP snooping drop legitimate requests until you marked the right port trusted? Did BPDU Guard or Root Guard trip on a port you didn't expect? Did the 'violation shutdown' port actually err-disable, and what did you have to type to bring it back?]*

## 12. Key Learning Outcomes

- I can explain the practical difference between standard, extended, and named ACLs — not just recite the syntax, but why each fits a different requirement.
- I understand why ACL rule order matters and can predict what a misordered ACL will actually do before testing it.
- I configured SSH-only management and can explain why Telnet and HTTP management are a real risk, not just a checklist item.
- I know the difference between BPDU Guard (protects access ports from rogue switches) and Root Guard (protects the topology from an unauthorized root election), including why they go on opposite ends of the same kind of link.
- I have a concrete example of DHCP Snooping breaking legitimate traffic until the trust boundary was configured correctly — not just that it can happen.

## 13. Conclusion

Security in this lab isn't one feature — it's several independent layers that each assume the others might fail. ACLs control who can reach what at Layer 3; Port Security and 802.1X-adjacent controls limit what's allowed to even join the network at Layer 2; DHCP Snooping and the guard features protect the control plane itself from being subverted. None of it is complicated in isolation, but getting all of it to coexist without breaking legitimate traffic — HR still needs the IT-Server, PCs still need DHCP, PortFast still needs to not undermine STP — is the actual skill this week was testing.

## Appendix A — Submission Checklist

| Deliverable                                            | Status                                                   |
|--------------------------------------------------------|----------------------------------------------------------|
| Cisco Packet Tracer (.pkt) project file                | [ Build using Section 9, then attach ]                   |
| GNS3 project file with all security configurations     | [ Build using Sections 4-7, then attach ]                |
| Network topology diagram (devices, VLANs, connections) | Included — Section 2, Figure 1                           |
| Configuration files for all routers and switches       | Complete — Section 7                                     |
| ACL verification (Standard, Extended, Named)           | [ Insert in Section 8.1 ]                                |
| SSH verification                                       | [ Insert in Section 8.2 ]                                |
| Layer 2 security verification                          | [ Insert in Section 8.3 ]                                |
| Connectivity test results                              | [ Insert in Section 8.4 ]                                |
| Troubleshooting report                                 | Template in Section 10 — replace with your real scenario |
| Final documentation (10+ pages)                        | This document                                            |
| LinkedIn Post                                          | [ See draft provided separately ]                        |
| GitHub Repository                                      | [ See README provided separately ]                       |